BrandON

Web-Based Banner Order Management Platform

Group-8 - Spring 2026

East West University

Dept. of Computer Science & Engineerin

CSE495

Project Management Report

Course Supervisor

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THE ELEVATOR PITCH

The Problem

- Offline banner shops create **physical friction** – customers must visit just to check prices
- No digital tracking; every status update is a manual phone call
- Lost orders, pricing inconsistency, and wasted customer time cost local printers real sales

The Solution

- A complete web platform enabling **24/7 online ordering** with zero physical visit required
- **Custom dynamic pricing** engine calculates costs in real time for any banner spec
- Secure **file uploads** + a full **admin dashboard** to manage every order lifecycle

Phase 1 – Initiating: Project Charter & Scope

✔ In Scope

- ✔ Public ordering pages
- ✔ Dynamic pricing calculation
- ✔ Admin CRUD dashboard
- ✔ Order status pipeline
- ✔ Authentication & RBAC

✘ Out of Scope

- ✘ Payment processing (Cash on Delivery only)
- ✘ Customer account registration
- ✘ Multi-store support

Executive Sponsor: Dr. Rezvi Shahriar — Department of CSE, East West University

Phase 1 – Initiating: Business Strategy

Value Proposition

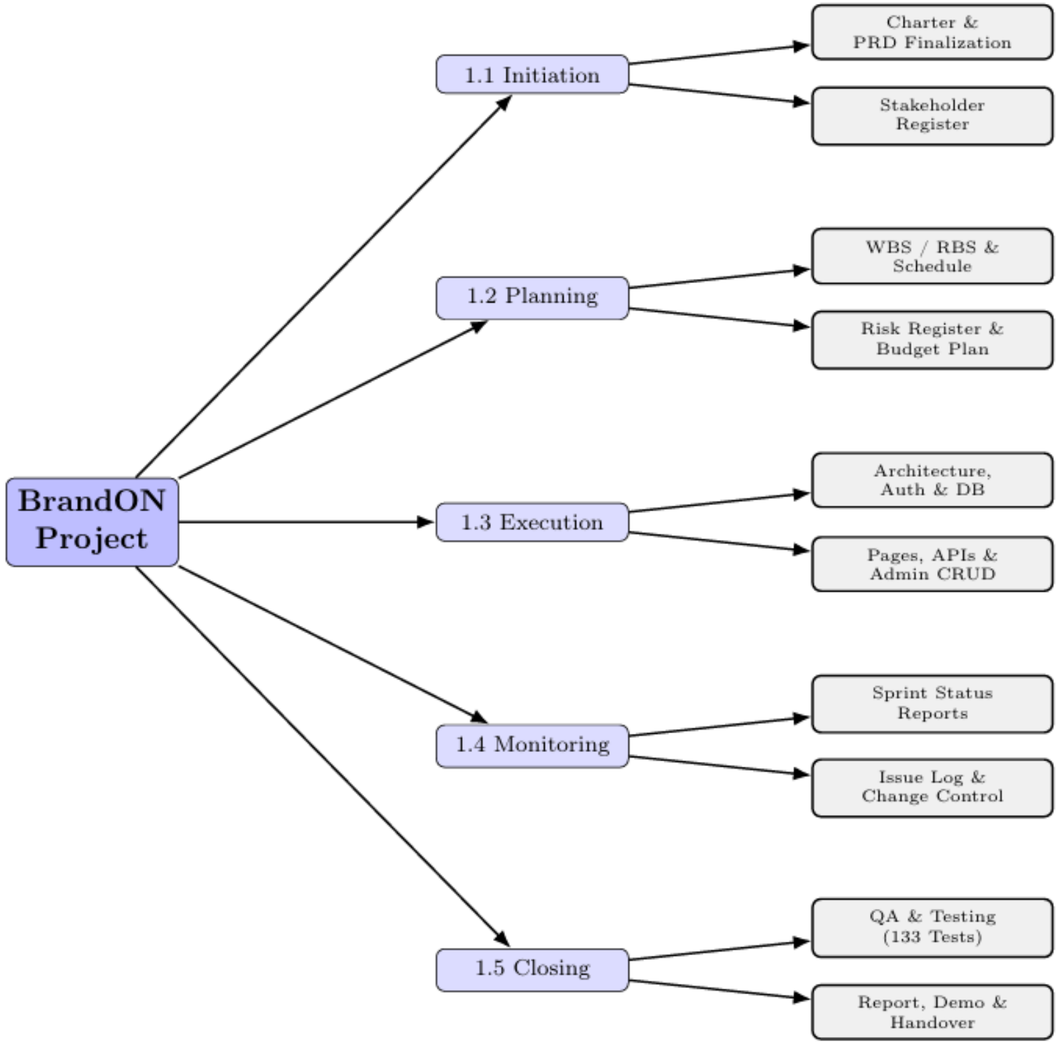
- **Instant digital pricing** for any banner specification
- **Paperless operations** for local SME print businesses
- Zero onboarding cost – browser-based, no installation

Market Feasibility (PESTEL)

- **Minimal startup cost** – fully open-source stack
- Growing SME demand for **digital order management**
- No regulatory barriers to web-based print commerce in BD
- Underserved niche with **immediate market fit**

Strategic Conclusion: Conditions are highly favourable. A low-cost web platform can immediately capture an underserved niche among local Bangladeshi print businesses.

Phase 2 – Planning: WBS & Resources



Team Structure & RACI

Lead Area	Member
Architecture	Mahamudur
Backend	Mahir
Frontend	Israt
QA & Testing	Hamim
Delivery / PM	Oishi

Phase 2 – Planning: Schedule Baseline

9-Sprint / 5-Month Delivery Roadmap

Sprints	Phase	Focus
1 – 2	Discovery & Infrastructure	Requirements, Docker, env setup
3 – 4	Architecture & Security	DB schema, RBAC middleware, auth
5 – 6	Core APIs & Admin	REST endpoints, pricing engine
7 – 8	Order Flow & Admin CRUD	Order pipeline, gallery, materials
9	QA & Docs	133 tests, PM report, handover

- **Baseline Budget:** BDT 7,000 – estimated via COCOMO model

Phase 3 – Executing: Sprint Deliverables

What Was Built – Feature Set

- ✔ Public banner **order form** with file upload
- ✔ **Dynamic pricing engine** (Standard, Bulk, Rush)
- ✔ Admin dashboard for **full order lifecycle**
- ✔ Gallery & materials **CRUD management**
- ✔ Contact message inbox for operators

Sprint Completion Against Baseline

Sprint(s)	Planned	Delivered
1 – 2	2	**2**
3 – 4	2	**2**
5 – 6	2	**2**
7 – 8	2	**2**
9	1	**1**

SPI = 1.00 – every sprint delivered on time.

Phase 3 – Executing: Stakeholder Communication

Communication Plan

- **Sprint Reviews** held at the end of every sprint with the supervisor
- **Weekly stand-ups** within the team to surface blockers early
- **Issue log** maintained to track and escalate unresolved items
- Sponsor informed of scope, budget, and timeline status at each milestone

Change Control Process

- All change requests assessed against the **scope baseline** before acceptance
- No out-of-scope features were added – **zero scope creep** recorded
- Schema changes required a **team review** before merging to maintain baseline integrity
- Final feature set matched the approved **Project Charter** exactly

Phase 4 – Monitoring & Controlling: Risk Management

Risk	Impact	Mitigation
Price manipulation via client-side tampering	**High**	Enforced server-side price recalculation; client values completely ignored.
Database schema churn	**Medium**	Strict Prisma migration review before every merge to `main`.
Team learning curve on Next.js / Docker	**Medium**	Sprints 1–4 dedicated to pair programming and knowledge sharing.

Phase 4 – Monitoring & Controlling: Quality & Costs

Quality Control – Automated Test Suite

133 Tests **0 Failures**

100% Pass Rate in 2.381 s

- **Unit (33):** Pricing logic & input validation
- **Integration (53):** Database CRUD & API contracts
- **System (24):** End-to-end user journeys
- **Regression (17):** Guards against re-introduced bugs

✔ Cost Control – Earned Value Management

31% Under Budget

Planned BDT 7,000 Actual BDT 4,800

- **CPI = 1.46** – high cost efficiency
- **SPI = 1.00** – perfectly on schedule
- **CV = +2,200 BDT** – positive cost variance

Phase 5 – Closing: Final Deliverables

✔ Handover Checklist

- ✔ Source Code (Git repository)
- ✔ Admin Credentials (documented securely)
- ✔ Docker Deployment Guide
- ✔ Full PM & Technical Documentation
- ✔ MVP Live Demo

MVP Feature Status

- ✔ Public ordering & file upload
- ✔ Dynamic pricing engine
- ✔ Admin CRUD dashboard
- ✔ Order status pipeline
- ✔ Secure admin authentication

All critical MVP features are functional and containerised for deployment.

Phase 5 – Closing: Lessons Learned & Sign-off

Key Lessons Learned

- **Early Charter clarity** prevented scope creep across all 9 sprints
- **Docker in Sprint 2** eliminated deployment environment discrepancies entirely
- **Server-side validation** is mandatory – client inputs must never be trusted for platform integrity

Thank You.